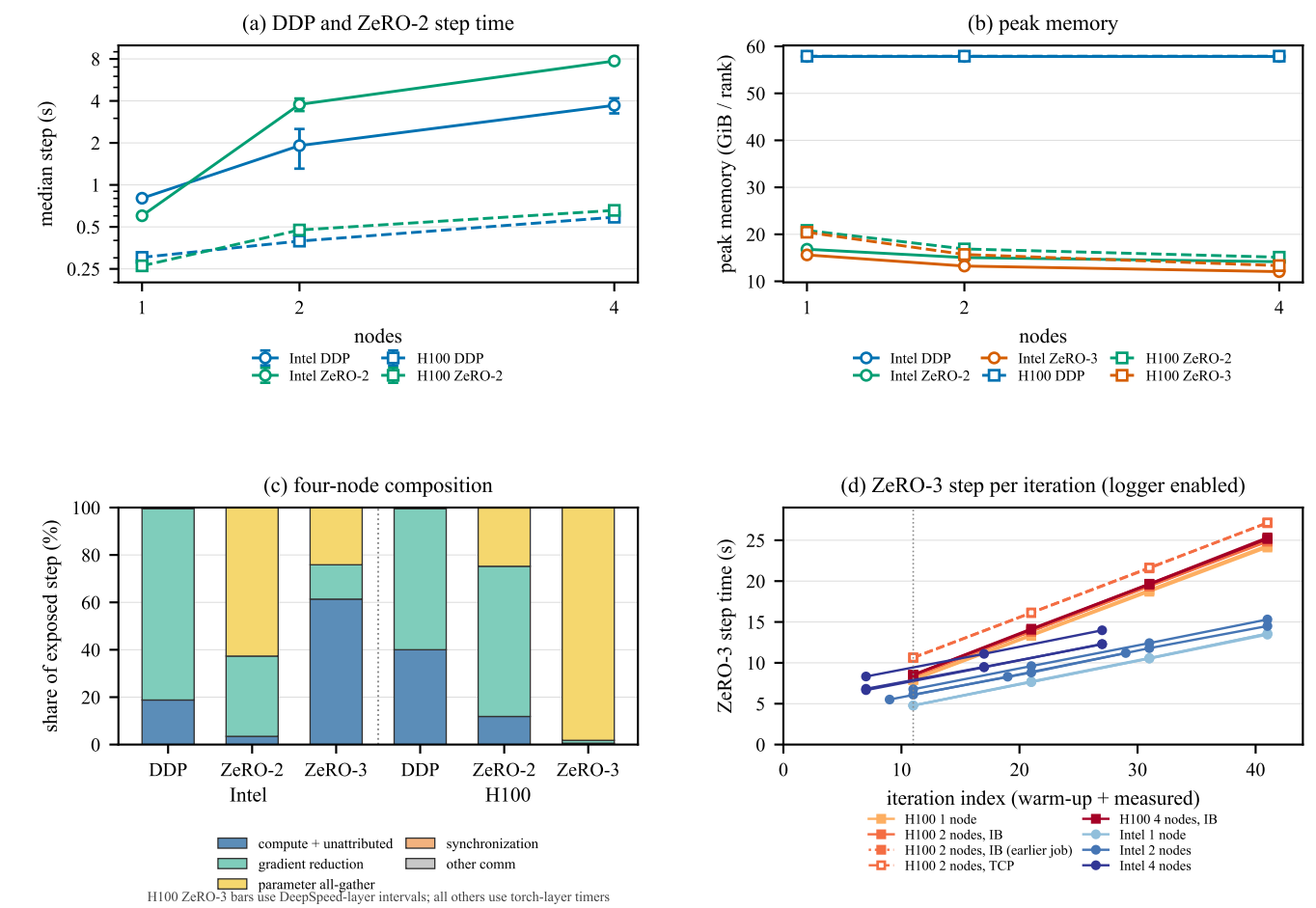
(a) DDP and ZeRO-2 step time
median step (s)
8
4
2
1
0.5
0.25
1
2
4
nodes
Intel DDP
Intel ZeRO-2
H100 DDP
H100 ZeRO-2
(b) peak memory
peak memory (GiB / rank)
60
50
40
30
20
10
1
2
4
nodes
Intel DDP
Intel ZeRO-2
Intel ZeRO-3
H100 DDP
H100 ZeRO-2
H100 ZeRO-3
(c) four-node composition
share of exposed step (%)
100
80
60
40
20
0
DDP
ZeRO-2
ZeRO-3
DDP
ZeRO-2
ZeRO-3
Intel
H100
compute + unattributed
gradient reduction
parameter all-gather
synchronization
other comm
H100 ZeRO-3 bars use DeepSpeed-layer intervals; all others use torch-layer timers
(d) ZeRO-3 step per iteration (logger enabled)
ZeRO-3 step time (s)
25
20
15
10
5
0
0
10
20
30
40
iteration index (warm-up + measured)
H100 1 node
H100 2 nodes, IB
H100 2 nodes, IB (earlier job)
H100 2 nodes, TCP
H100 4 nodes, IB
Intel 1 node
Intel 2 nodes
Intel 4 nodes